PLANNING: Conceitos Fundamentais

Gustavo Giménez-Lugo

PLANEJAMENTO: Conceitos Fundamentais

Gustavo Giménez-Lugo

Existe uma Representação do Mundo na “cabeça” ?

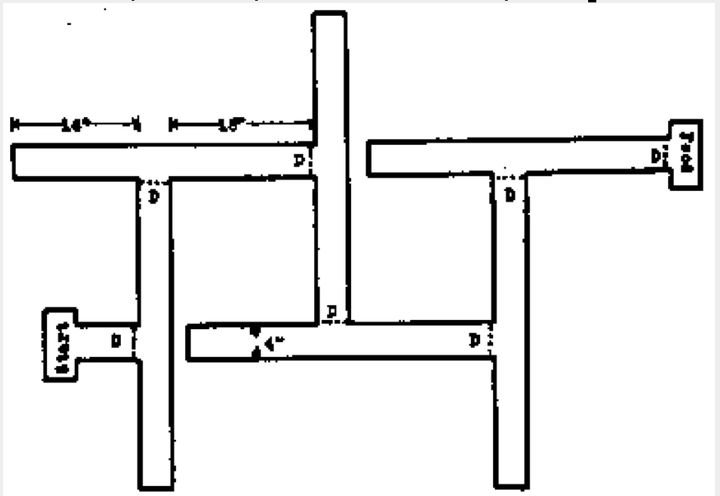
É tudo repetição e ação-reação para aprender?

Existe uma Representação do Mundo na “cabeça” ?

É tudo repetição e ação-reação para aprender?

1- Pode existir uma Representação do Mundo na “cabeça” ? [Mapas Cognitivos em Ratos, Tolman, Ritchie & Kalish, 1946]





6-Unit Alley T-Maze

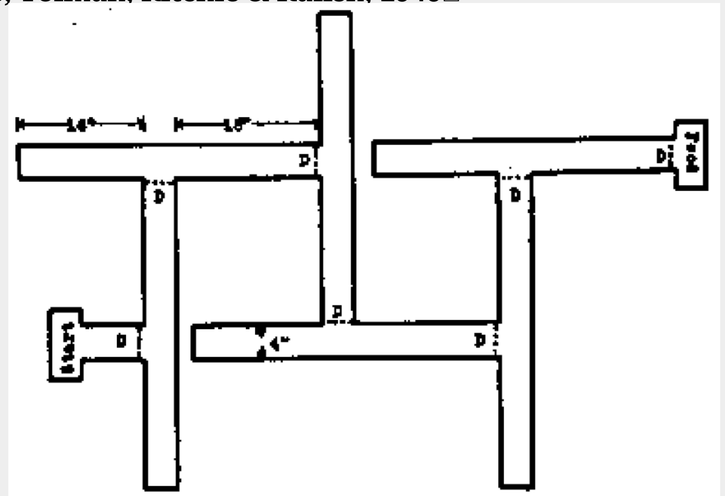
FIG. 4

(From H. C. Blodgett, The effect of the introduction of reward upon the maze performance of rats. *Univ. Calif. Publ. Psychol.*, 1929, 4, No. 8, p. 117.)

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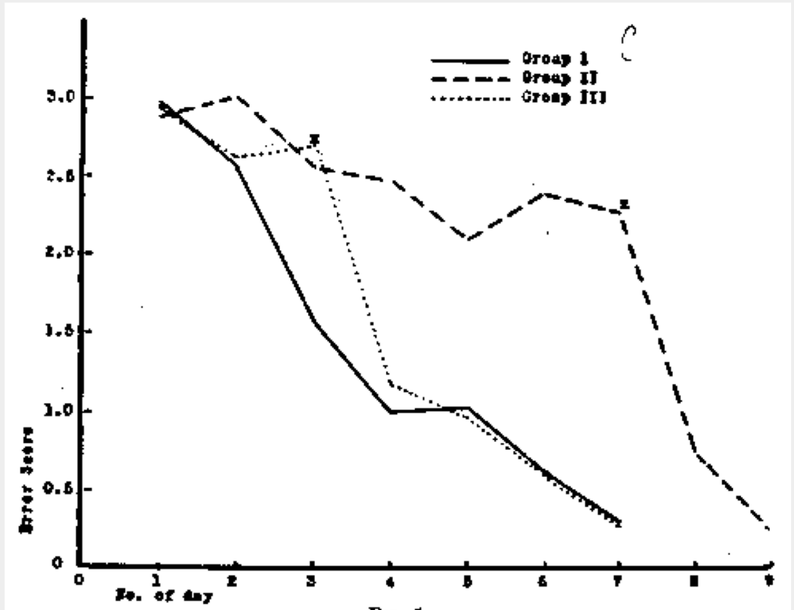


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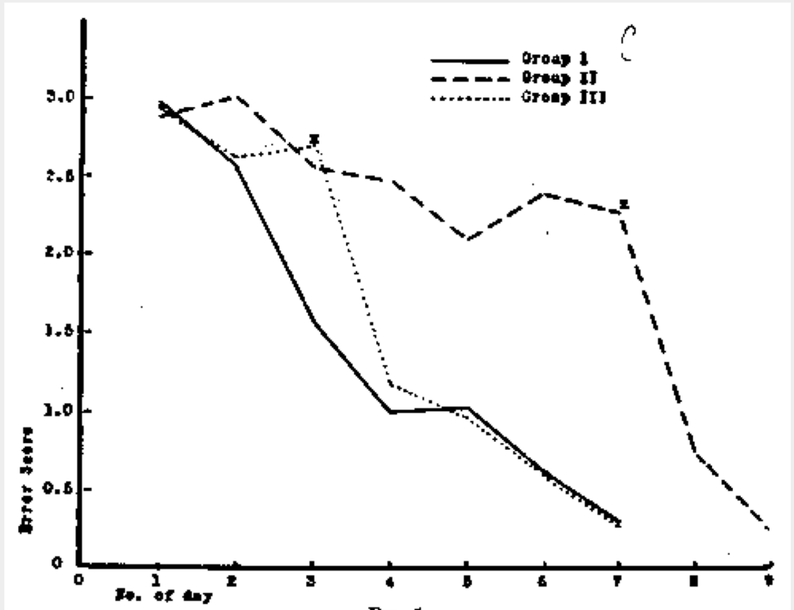


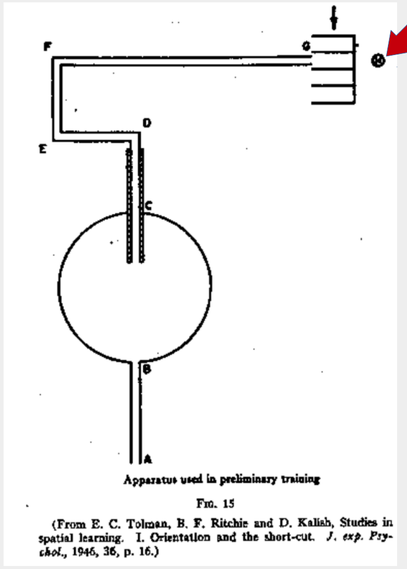
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Apparatus used in preliminary training

FIG. 15

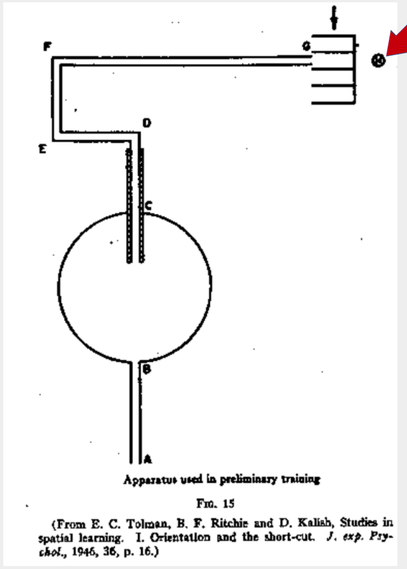
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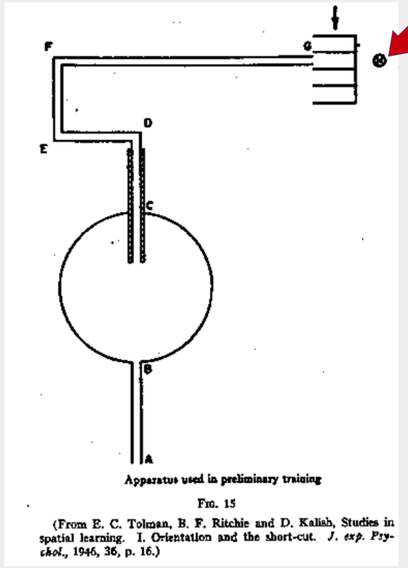
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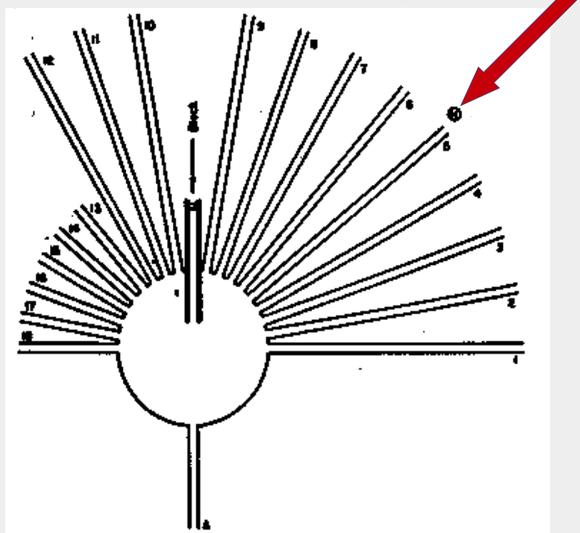
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Apparatus used in the test trial

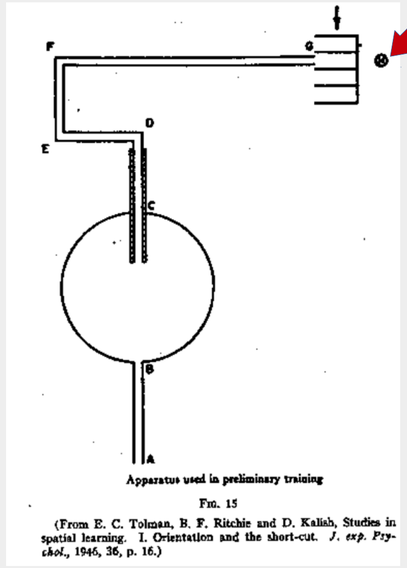
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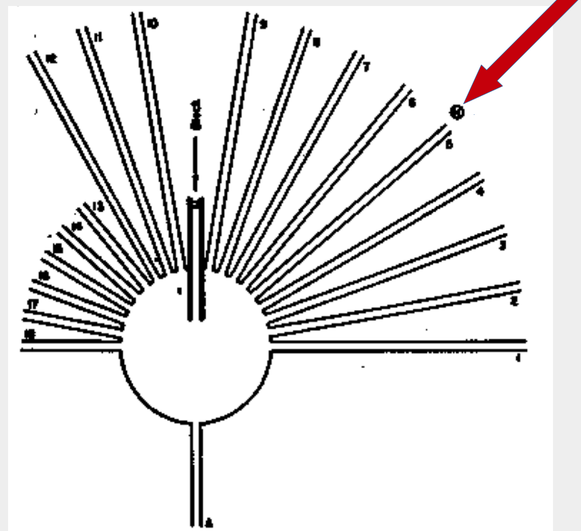
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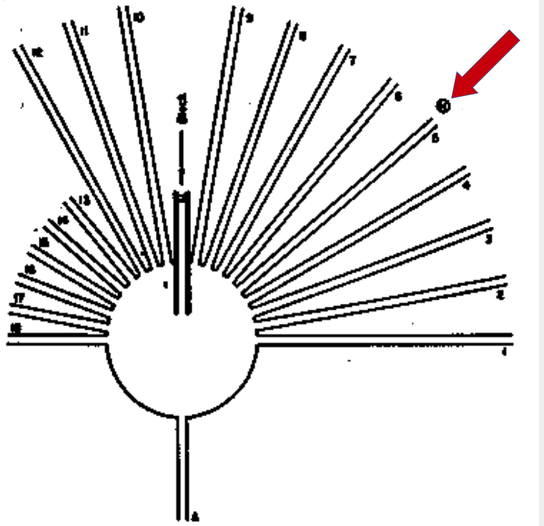
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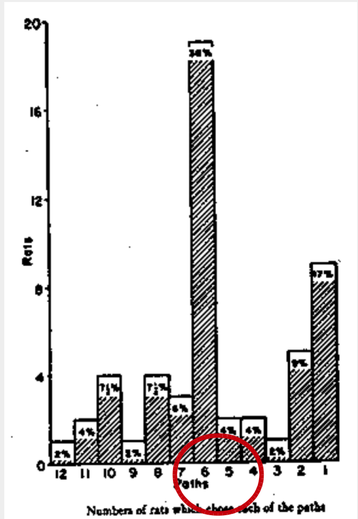


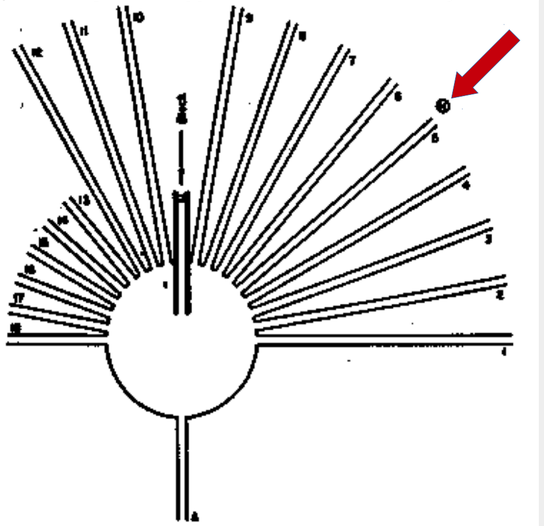
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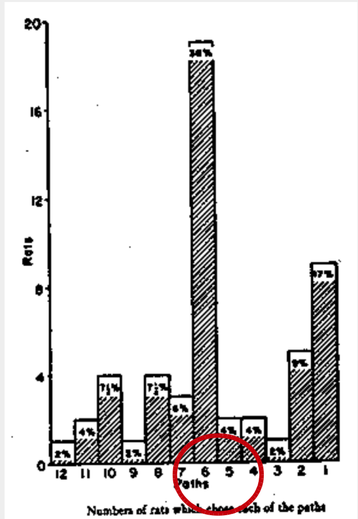


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2- Pode-se usar a Representação para IMAGINAR/PROJETAR/"PREVER" o que pode acontecer no mundo OBSERVADO?

Funt, B.V., 'Problem-Solving with Diagrammatic Representations', Artificial Intelligence, Vol. 13 No. 4 1980, pp. 201-230. https://www.cs.sfu.ca/~funt/ProblemSolvingWithDiagrammaticRepresentations_AIJournal1980.pdf

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2- Pode-se usar a Representação para IMAGINAR/PROJETAR/"PREVER" o que pode acontecer no mundo OBSERVADO?

(WHISPER, a high level reasoner, computer problem solving testing stability of a “BLOCKS WORLD”)

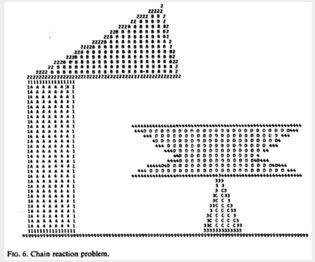


Fig. 6. Chain reaction problem.

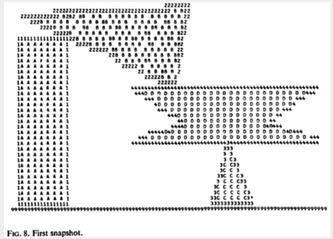


Fig. 8. First snapshot.

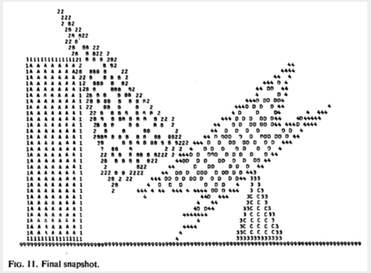


Fig. 11. Final snapshot.

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(WHISPER, um raciocinador de alto nível, solução de problemas de computador, testando a estabilidade de um “MUNDO DE BLOCOS”)

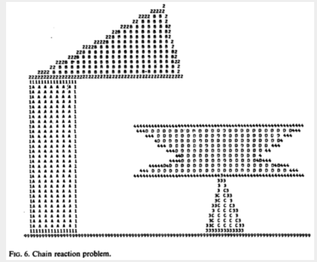


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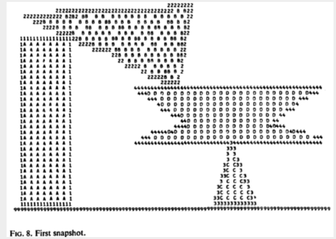


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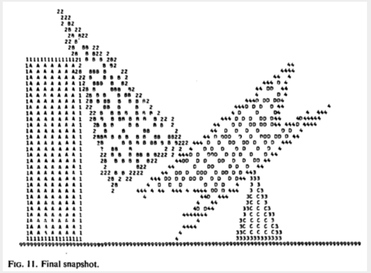


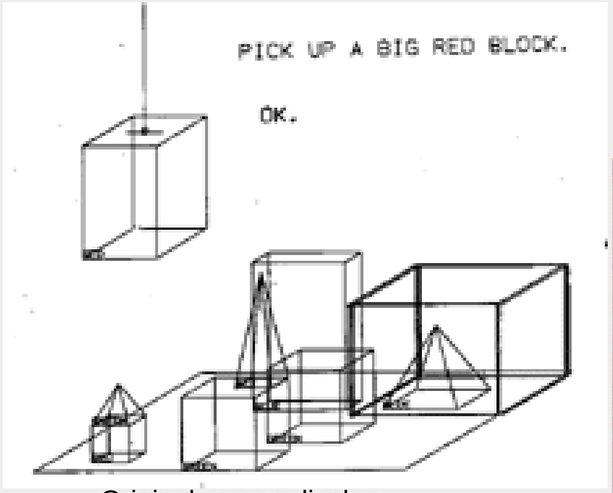
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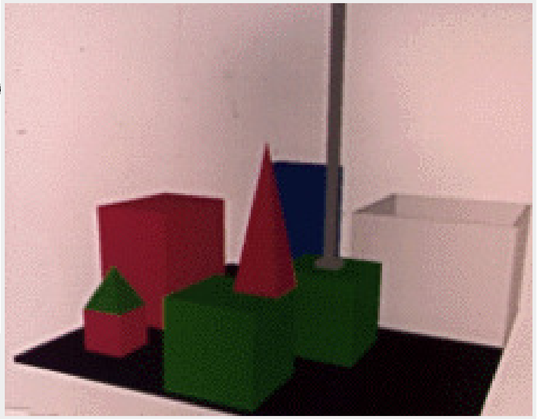
3- Pode-se IMAGINAR o que pode acontecer se EU FAÇO ALGO no MUNDO ?

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3- Pode-se IMAGINAR o que pode acontecer se EU FAÇO ALGO no MUNDO ? (SHRDLU)



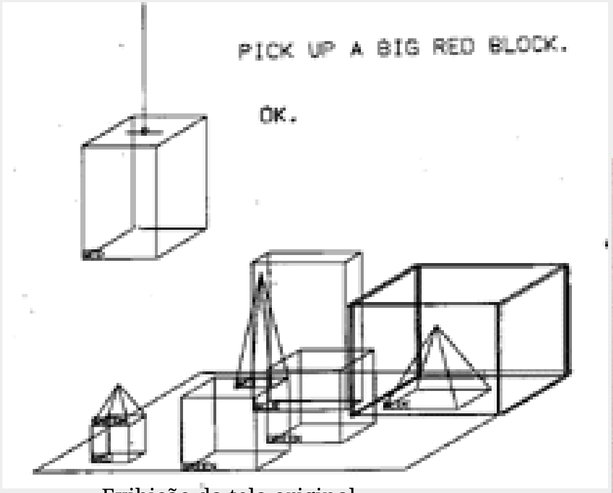
Original screen display



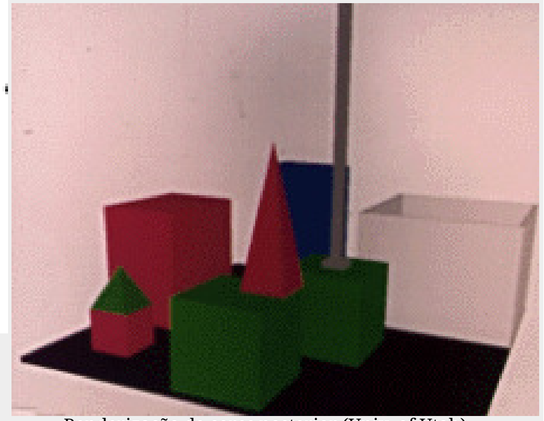
Later color rendering (Univ. of Utah)

SOURCE:
Terry Winograd's dissertation, which was issued as MIT AI Technical Report 235, February 1971 with the title Procedures as a Representation for Data in a Computer Program for Understanding Natural Language. Published as a full issue of the journal Cognitive Psychology Vol. 3 No 1, 1972, and as a book, Understanding Natural Language (Academic Press, 1972).
http://hci.stanford.edu/winograd/shrdlu/

3- Pode-se IMAGINAR o que pode acontecer se EU FAÇO ALGO no MUNDO ? (SHRDLU)



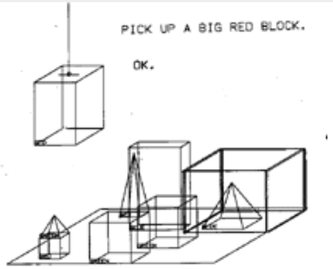
Exibição da tela original

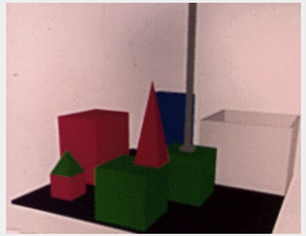


Renderização de cores posterior (Univ. of Utah)

FONTE:
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•Stanford •edu/winograd/shrdlu/

3- Pode-se IMAGINAR o que pode acontecer se EU FAÇO ALGO no MUNDO ? (SHRDLU, 1968-1970 a program for understanding natural language)



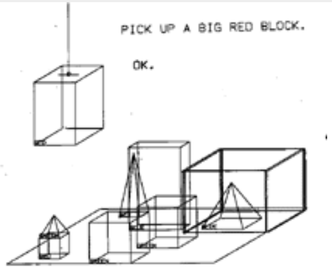


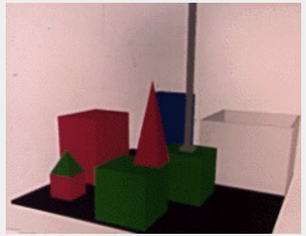
The dialog that was used as a SHRDLU demo:

Person: PICK UP A BIG RED BLOCK.
Computer: OK. (does it)
Person: GRASP THE PYRAMID.
Computer: I DON'T UNDERSTAND WHICH PYRAMID YOU MEAN.
Person: FIND A BLOCK WHICH IS TALLER THAN THE ONE YOU ARE HOLDING AND PUT IT INTO THE BO
Computer: BY "IT", I ASSUME YOU MEAN THE BLOCK WHICH IS TALLER THAN THE ONE I AM HOLDING.
Computer: OK. (does it)
Person: WHAT DOES THE BOX CONTAIN?
Computer: THE BLUE PYRAMID AND THE BLUE BLOCK.
Person: WHAT IS THE PYRAMID SUPPORTED BY?
Computer: THE BOX.
Person: HOW MANY BLOCKS ARE NOT IN THE BOX?
Computer: FOUR OF THEM.
Person: IS AT LEAST ONE OF THEM NARROWER THAN THE ONE WHICH I TOLD YOU TO PICK UP?
Computer: YES, THE RED CUBE.
Person: IS IT SUPPORTED?
Computer: YES, BY THE TABLE.
Person: CAN THE TABLE PICK UP BLOCKS?
Computer: NO.
Person: CAN A PYRAMID BE SUPPORTED BY A BLOCK?

SOURCE:
SHRDLU is described in Terry Winograd's dissertation, which was issued as MIT AI Technical Report 235, February 1971 with the title Procedures as a Representation for Data in a Computer Program for Understanding Natural Language It was published as a full issue of the journal Cognitive Psychology Vol. 3 No 1, 1972, and as a book, Understanding Natural Language (Academic Press, 1972).
http://hci.stanford.edu/winograd/shrdlu/

3- Pode-se IMAGINAR o que pode acontecer se EU FAÇO ALGO no MUNDO ? (SHRDLU, 19681970 - um programa para compreensão da linguagem natural)





The dialog that was used as a SHRDLU demo:

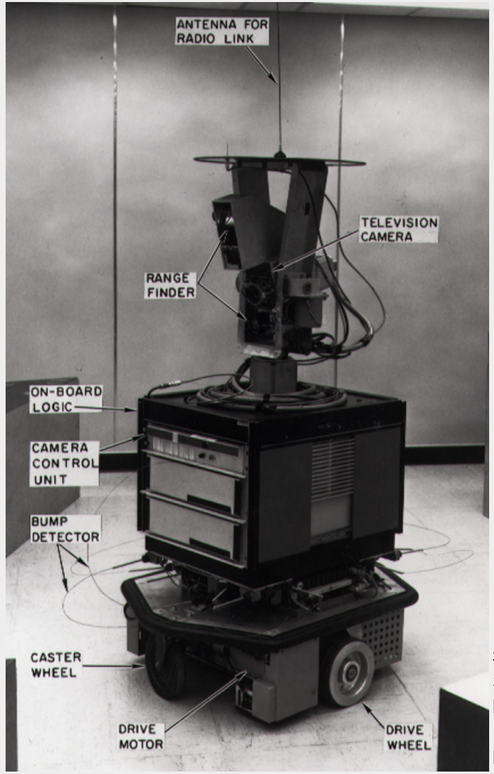
Person: PICK UP A BIG RED BLOCK.
Computer: OK. (does it)
Person: GRASP THE PYRAMID.
Computer: I DON'T UNDERSTAND WHICH PYRAMID YOU MEAN.
Person: FIND A BLOCK WHICH IS TALLER THAN THE ONE YOU ARE HOLDING AND PUT IT INTO THE BO
Computer: BY "IT", I ASSUME YOU MEAN THE BLOCK WHICH IS TALLER THAN THE ONE I AM HOLDING.
Computer: OK. (does it)
Person: WHAT DOES THE BOX CONTAIN?
Computer: THE BLUE PYRAMID AND THE BLUE BLOCK.
Person: WHAT IS THE PYRAMID SUPPORTED BY?
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FONTE:
SHRDLU é descrito na dissertação de Terry Winograd, publicada como MIT AI Technical Report 235, fevereiro de 1971 com o título Procedures as a Representation for Data in a Computer Program for Understanding Natural Language. Foi publicada como uma edição completa da revista Cognitive Psychology Vol. 3 No 1, 1972, e como livro, Understanding Natural Language (Academic Press, 1972).
http://hci.stanford.edu/winograd/shrdlu/

4- Pode-se IMAGINAR e depois FAZER algo no MUNDO … e depois ainda atualizar crenças sobre o MUNDO ?

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4- Pode-se IMAGINAR e depois FAZER algo no MUNDO … e depois ainda atualizar crenças sobre o MUNDO ? (SHAKEY, 1966-1972)



"SHAKEY was an important development because the RESEARCH concentrated on WHAT the ROBOT THOUGHT and NOT on WHAT it DID" (01m00s)

"SHAKEY foi um desenvolvimento importante porque a PESQUISA estava interessada NO QUE O ROBÔ PENSAVA, NÃO no que FAZIA" (01m00s)

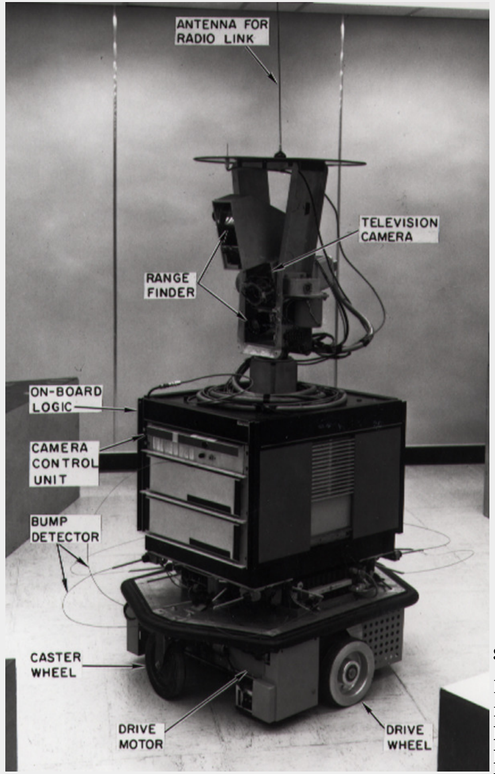
Shakey The Robot
https://vimeo.com/5072714

[Shakey was developed at the Artificial Intelligence laboratory of the Stanford Research Institute in 1966 through 1972 with Charles Rosen as project manager. Major contributors included Nils Nilsson, Alfred Brain, Bertram Raphael, Richard Duda, Peter Hart, Richard Fikes, Richard Waldinger, Thomas Garvey, Jay Tenenbaum, and Michael Wilber. Programming was primarily in LISP.

Story by Scott McGrew. Directed by Barton Bishoff. Additional photography by Donald Hardy and John Chiala.]

Shakey em 1972
SOURCE:
https://en.wikipedia.org/wiki/Shakey_the_robot
#/media/File:SRI_Shakey_with_callouts.jpg
https://vimeo.com/5072714
https://www.youtube.com/watch?v=7bsEN8mwUB8

4- Pode-se IMAGINAR e depois FAZER algo no MUNDO … e depois ainda atualizar sobre o MUNDO ? (SHAKEY, 1966-1972)



"SHAKEY foi um desenvolvimento importante porque a PESQUISA se concentrou no QUE O ROBÔ PENSOU e NÃO NO QUE FEZ" (01m00s)

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Shakey, o robô
https://vimeo.com/5072714

Shakey foi desenvolvido no laboratório de Inteligência Artificial do Stanford Research Institute de 1966 a 1972, com Charles Rosen como gerente de projeto. Os principais contribuidores incluíram Nils Nilsson, Alfred Brain, Bertram Raphael, Richard Duda, Peter Hart, Richard Fikes, Richard Waldinger, Thomas Garvey, Jay Tenenbaum e Michael Wilber. A programação era principalmente em LISP.

História de Scott McGrew. Dirigido por Barton Bishoff. Fotografia adicional de Donald Hardy e John Chiala.

Shakey em 1972

FONTE: https://en.wikipedia.org/wiki/Shakey□o□
robô #/media/File:SRI□Shakey□com□callouts.jpg
https://vimeo.com/5072714
https://www.youtube.com/watch?v=7
bsEN8mwUB8